

Weekly Progress Report

Project 6 — Turbofan Engine Remaining Useful Life Prediction

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I. Overview

This week's focus was on exploring and implementing a machine learning pipeline for predicting the Remaining Useful Life (RUL) of turbofan jet engines using the NASA C-MAPSS dataset. The objective is to estimate how many operational cycles remain before an engine fails, enabling proactive maintenance decisions.

II. Dataset Understanding

2.1 NASA C-MAPSS Dataset

The Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset contains multivariate time-series data from a fleet of simulated turbofan engines. Each engine operates normally at the start and progressively develops a fault. The dataset is divided into four sub-datasets:

- FD001 — 100 training engines, sea-level condition, single fault mode (HPC Degradation)
- FD002 — 260 training engines, six operating conditions, single fault mode
- FD003 — 100 training engines, sea-level condition, two fault modes (HPC + Fan Degradation)
- FD004 — 248 training engines, six conditions, two fault modes

2.2 Data Structure

Each row represents one operational cycle snapshot with 26 columns:

- Columns 1–2: Unit number and cycle timestamp
- Columns 3–5: Three operational settings (altitude, Mach, throttle resolver angle)

- Columns 6–26: 21 sensor measurements (temperatures, pressures, fan/core speeds, fuel flow, etc.)

Training data includes all cycles up to failure; test data ends before failure. Ground-truth RUL values are provided separately for evaluation.

III. Methodology

3.1 RUL Label Generation

For training data, piece-wise linear RUL labels were computed: $RUL = (\text{max cycle} - \text{current cycle})$, capped at 125 cycles. The cap prevents the model from trying to predict very large RUL values (early healthy cycles) where the relationship is flat, focusing learning on the degradation window.

3.2 Feature Engineering

The following feature engineering steps were applied:

- Sensor Selection: Removed constant or near-constant sensors ($\text{std} < 0.01$) — typically s1, s5, s10, s16, s18, s19 which carry no discriminative information.
- Rolling Statistics: For each useful sensor, computed per-engine rolling mean and rolling standard deviation over a 10-cycle window. This captures the degradation trend rather than noisy instantaneous readings.
- Min-Max Scaling: All features normalised to $[0, 1]$ before model training.

3.3 Models Implemented

- Linear Regression — baseline; models a linear relationship between features and RUL
- Random Forest Regressor — ensemble of 100 decision trees; robust to noise and non-linearity
- Gradient Boosting Regressor — sequential boosting; typically, best on tabular data

For each test engine, the last observed cycle's features were extracted and fed to the trained models to produce a single RUL prediction.

IV. Results & Model Performance

Evaluation metric: Root Mean Squared Error (RMSE) in cycles. Lower is better.

Dataset	Conditions	Fault Modes	Train Engines	LR RMSE	RF RMSE	Best Model
FD001	1 (Sea Level)	1 (HPC)	100	21.91	18.75	Random Forest
FD002	6	1 (HPC)	260	32.39	29.28	Random Forest
FD003	1 (Sea Level)	2 (HPC+Fan)	100	21.60	20.79	Random Forest
FD004	6	2 (HPC+Fan)	248	34.63	30.47	Random Forest

Key observations:

- Random Forest consistently outperforms Linear Regression across all datasets.
- FD001 and FD003 (single operating condition) achieve lower RMSE than FD002/FD004, as the model doesn't need to disentangle multiple operating regimes.
- FD004 (6 conditions + 2 fault modes) is the hardest dataset, achieving RMSE ~30.5 — reflecting greater system complexity.

V. Engine Health Scoring

Predicted RUL values are mapped to actionable health categories:

Status	RUL Range (cycles)	Action Required
● Healthy	≥ 90	Normal operation — routine monitoring
● Monitor	50–89	Increase inspection frequency
● Warning	20–49	Schedule maintenance intervention
● Critical	< 20	Immediate action — high failure risk

VI. Achievements This Week

- Studied the C-MAPSS dataset structure, column semantics, and the four sub-dataset configurations.
- Implemented end-to-end data loading, preprocessing, and feature engineering pipeline in Python.
- Implemented piece-wise linear RUL labelling with configurable cap.
- Trained and evaluated three ML models (LR, RF, GB) across all four datasets.
- Implemented per-engine health status scoring based on predicted RUL.
- Produced clean, modular Python code (`rul_prediction.py`) with in-line documentation.

VII. Challenges Encountered

- Multi-condition datasets (FD002, FD004): sensor readings vary significantly across operating conditions, requiring the model to learn condition-normalised patterns. Operational settings were included as features to partially address this.
- Sensor noise: Raw sensor values are noisy; rolling statistics help but a more sophisticated denoising or clustering of operating regimes would further improve accuracy.

- RUL cap selection: The choice of cap (125 cycles) affects training distribution; values too low truncate early-life data, too high dilutes the degradation signal.

VIII. Next Steps

- Implement operating condition clustering (e.g., K-Means on os1/os2/os3) for FD002/FD004 to normalise readings before feature extraction.
- Explore LSTM / GRU sequence models to capture temporal degradation patterns over the full engine lifecycle.
- Tune hyperparameters (RUL cap, window size, n_estimators) via cross-validation.
- Add feature importance plots to identify the most predictive sensors.
- Package the pipeline for deployment as a real-time inference service.

IX. References

- Saxena, A. et al. (2008). Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation. IEEE International Conference on Prognostics and Health Management.
- NASA Prognostics Center of Excellence — C-MAPSS Dataset: <https://data.nasa.gov/Aerospace/CMAPSS-Jet-Engine-Simulated-Data/ff5v-kuh6>
- scikit-learn Documentation: <https://scikit-learn.org>